

PolyAgora V7.4b

Soft Local-Star Manifold Upgrade

PolygonEye / PolyAgora CTO Specification

May 10, 2026

1 V7.4b — Soft Local-Star Manifold Upgrade

The first implementation of Strategy Manifold V1 used a hard Local-Star masking rule:

$$\mathbf{1}_{\phi_i \in B_t^*}.$$

This architecture proved excessively brittle. Strategies outside the selected Local Star were completely suppressed, producing over-concentration, reduced diversification, and unstable runtime behavior. Empirically, this caused degradation relative to the V7.3 mixture-template architecture.

The purpose of V7.4b is therefore to preserve the Strategy Manifold framework while replacing hard Local-Star selection with a soft multi-block aggregation layer.

1.1 Block Scores

For each block B_m , compute:

$$Q_m(t) = \mu_m(X_t) \cdot \Gamma_t(B_m),$$

where:

$\mu_m(X_t) \in [0, 1]$	soft membership of runtime state X_t ,
$\Gamma_t(B_m) = \min_{\phi_i, \phi_j \in B_m} W_t(i, j)$	internal survival coherence.

1.2 Top-K Local-Star Selection

Instead of selecting only:

$$B_t^* = \arg \max_m Q_m(t),$$

define the Top-K admissible block set:

$$\mathcal{B}_t^{(K)} = \{B_{(1)}, B_{(2)}, \dots, B_{(K)}\},$$

where blocks are ordered by descending $Q_m(t)$.

Default:

$$K = 2.$$

Recommended testing:

$$K \in \{2, 3\}.$$

1.3 Soft Block Weights

Define soft block aggregation weights:

$$\rho_m(t) = \frac{\exp(\tau Q_m(t))}{\sum_{\ell \in \mathcal{B}_t^{(K)}} \exp(\tau Q_\ell(t))},$$

where:

$$\tau > 0$$

is a concentration parameter.

Interpretation:

- low τ produces smoother diversification,
- high τ approaches hard Local-Star selection.

Default:

$$\tau = 4.$$

Recommended testing:

$$\tau \in \{2, 4, 6\}.$$

1.4 Soft Strategy Membership

Define strategy-level manifold membership:

$$M_i(t) = \sum_{m \in \mathcal{B}_t^{(K)}} \rho_m(t) \cdot \mathbf{1}_{\phi_i \in B_m}.$$

This replaces the previous hard mask:

$$\mathbf{1}_{\phi_i \in B_t^*}.$$

Interpretation:

Strategies may partially participate across multiple coherent blocks simultaneously, preserving buffer behavior and reducing regime discontinuity.

1.5 Final Strategy Score

The runtime strategy score becomes:

$$p_i(t) = A_i(t) \cdot q_i(t) \cdot M_i(t) \cdot G_i(D_t),$$

where:

$$\begin{aligned} A_i(t) &= \text{strategy admissibility,} \\ q_i(t) &= \text{internal quality score,} \\ M_i(t) &= \text{soft manifold membership,} \\ G_i(D_t) &= \text{Driver Seat deformation multiplier.} \end{aligned}$$

1.6 Normalization

Normalize:

$$w_i^{raw}(t) = \frac{p_i(t)}{\sum_j p_j(t)}.$$

Apply zone scaling:

$$w_i(t) = g(Z_t) \cdot w_i^{raw}(t).$$

Cash allocation:

$$w_{\text{cash}}(t) = 1 - \sum_i w_i(t).$$

1.7 Dynamic Survival Weighting

The runtime survival matrix remains:

$$W_t = \lambda S_0 + (1 - \lambda) R_t.$$

Recommended parameter testing:

$$\lambda \in \{0.50, 0.65, 0.80\}.$$

Interpretation:

- high λ emphasizes prior structural assumptions,
- low λ increases responsiveness to rolling survivability.

1.8 Block Construction Threshold

Blocks should be constructed from survivability coherence graphs:

$$G_t = (\Phi, E_t),$$

where:

$$(i, j) \in E_t \iff W_t(i, j) \geq \theta_S.$$

Recommended testing:

$$\theta_S \in \{0.45, 0.55, 0.65\}.$$

Blocks may initially be implemented as:

- connected components,
- cliques,
- or community-detection clusters.

1.9 Expected Improvement over V7.4

V7.4 suffered from excessive exclusivity due to hard Local-Star masking.

V7.4b is expected to:

- restore diversification,
- preserve structural coherence,
- improve Sharpe ratio,
- reduce regime discontinuity,
- preserve buffer behavior,
- avoid single-block dominance,
- improve transition stability.

1.10 Validation Criteria

Success criteria for V7.4b:

1. Sharpe ratio recovers toward or exceeds V7.3,
2. Max drawdown remains below Momentum and Equal Weight,
3. Volatility does not materially expand,
4. No single block dominates more than 80% of timesteps,
5. Turnover remains operationally realistic,
6. Zone transitions become smoother and structurally interpretable.

1.11 Core Design Rule

Do not abandon Strategy Manifold V1.

The degradation in V7.4 was caused by excessive Local-Star exclusivity, not by failure of the strategy-eigenfield architecture itself.

V7.4b therefore preserves:

PolyAgora = Runtime Governance of Strategy Eigenfields,

while replacing hard Local-Star selection with soft manifold aggregation.